# Jonathan Tybirk

Copenhagen, Denmark

+45 50 59 22 96
jonathantybirk@gmail.com

jonathantybirk.com
github.com/jonathantybirk
linkedin.com/in/jonathantybirk


## Education

- **Technical University of Denmark (DTU)** *Sep 2023 – Dec 2026 (expected)*
  *BSc Artificial Intelligence and Data* GPA: 10.7 / 12
  - 1st place, Danish National Student Championship in AI 2025
  - Summer exchange, Nanyang Technological University (NTU), Singapore (2025)

## Research and Projects

- **Equivariant GNNs for Center-of-Mass Prediction in LiDAR Point Clouds** *Feb 2025 – Jun 2025*
  *SE(3)-equivariant GNN implementation and empirical evaluation* PyTorch, e3nn
  - Built a reproducible experimental pipeline: synthetic point-cloud generation, training, and evaluation
  - Implemented an SE(3)-equivariant GNN in PyTorch and benchmarked against geometric baselines and data-augmented non-equivariant GNNs
  - Matched the data-augmented GNN while enforcing exact rotation/translation equivariance; analyzed accuracy vs inference-time tradeoffs

- **Offline Medical Topic Identification and Statement Verification** *Aug 2025*
  *Healthcare RAG subchallenge of the Danish National Student Championship in AI 2025* Python, offline RAG
  - Winning solution, achieving 91% evaluation accuracy; team also placed 1st overall
  - Implemented BM25 indexing and embedding search over a 115-topic corpus, tuned via Bayesian optimization; used a local LLM to verify claims against retrieved evidence

## Experience

- **Hive Robots** *Jan 2026 – Present*
  *Student Robotics Engineer* Copenhagen, Denmark
  - Reproducing and extending vision-language-action (VLA) robotics methods; building baselines and evaluation pipelines
  - Generated teleoperation training data and fine-tuned an Isaac GR00T-based VLA manipulation policy; integrated with a humanoid robot navigation stack

- **Netcompany** *May 2025 – Jan 2026*
  *Student Software Developer* Copenhagen, Denmark
  - Designed and shipped an end-to-end invoicing report export feature for Denmark's national payment system; demoed to government stakeholders

- **Danish Student Association for Rocketry (DanSTAR)** *Sep 2024 – Present*
  *Software Team Member* Lyngby, Denmark
  - Implemented a TCP protocol between Mission Control and the STM32 flight computer; resolved persistent connection failures
  - Developed embedded telemetry components for the flight computer (C++)

- **Technical University of Denmark (DTU)** *Jan 2024 – May 2025*
  *Teaching Assistant* Lyngby, Denmark
  - Delivered weekly lectures and live programming sessions for the *Computer Programming* course for 120 students, including exercise walkthroughs and in-class Q&A
  - Led 2 additional TAs and supervised student projects in Deep Q-learning, ResNets, and text embeddings

## Technical Skills

**Languages**: Python, C++, Java, SQL
**ML**: PyTorch; geometric deep learning, vision transformers, deep RL
**Tools**: Linux, Git, Docker, GitHub Actions, ROS 2, HPC
**Frameworks**: FastAPI, Spring Boot, React

## Leadership & Activities

- **Treasurer and Board Member,** Danish Student Association for Rocketry (DanSTAR) *Feb 2025 – Present*